LA(1)LR(0)

임기정

잇창명님께 감사드립니다. 다음은 LALR 파서를 생성하는 데에 필요한 정의이다.

(i) Let $G \triangleq (N, T, P, S)$ be a CFG.
(ii) $G' \triangleq (N', T', P', S') = (N \uplus \{S'\}, T \uplus \{\$\}, P \uplus \{[S' \to S]\}, S')$.
(iii) $V \triangleq N \uplus T$. $V' \triangleq N' \uplus T'$.
(iv) $L(G) := \{z \in T^* \mid S \Rightarrow_P^* z\}$. $L(G') := \left\{z \in T^* \,\middle|\, S' \underset{\mathbf{rm}\ P'}{\Longrightarrow}^+ z\right\}$. $L(G) = L(G')$.
(v) $I \triangleq \{[A \to \alpha \cdot \beta] \mid [A \to \alpha\beta] \in P'\}$.
(vi) $\mathrm{Cl}(q) =_\mu q \cup \{[A \to \cdot\omega] \mid [A \to \omega] \in P', [B \to \alpha_1 \cdot A\alpha_2] \in \mathrm{Cl}(q)\}$ for $q \in \mathcal{P}(I)$.
(vii) $q_0 \triangleq \mathrm{Cl}(\{[S' \to \cdot S]\})$.
(viii) $\mathrm{GOTO}(q, X) \triangleq \mathrm{Cl}(\{[A \to \alpha X \cdot \beta] \mid [A \to \alpha \cdot X\beta] \in q\})$ for $q \in \mathcal{P}(I)$ and $X \in V$.
(ix) $Q \triangleq \mathrm{PT} \setminus \{\emptyset\}$ where $\mathrm{PT} =_\mu \{q_0\} \cup \{\mathrm{GOTO}(q, X) \mid q \in \mathrm{PT}, X \in V\}$.
(x) $\varepsilon : p \to q \overset{\text{def}}{\Longleftrightarrow} p \in Q \wedge p = q$. $X\alpha : p \to q \overset{\text{def}}{\Longleftrightarrow} p \in Q \wedge \alpha : \mathrm{GOTO}(p, X) \to q$.
(xi) $\mathrm{next}(q, X) := \mathrm{GOTO}(q, X)$ for $q \in Q$ and $X \in V$.
(xii) $\mathrm{reduce}(q, t) := \{[A \to \omega] \mid [A \to \omega\cdot] \in q, t \in \mathbf{LA}(q, [A \to \omega])\}$ for $q \in Q$ and $t \in T'$.
(xiii)

$$\frac{q'' = \mathrm{next}(q', t) \quad q'' \neq \emptyset}{(\alpha : q \to q', tz) \vdash (\alpha t : q \to q'', z)}\ \text{Shift}$$

(xiv)

$$\frac{[A \to \omega] \in \mathrm{reduce}(q', t) \quad q'' = \mathrm{next}(q', A) \quad q'' \neq \emptyset}{(\alpha\omega : q \to q', tz) \vdash (\alpha A : q \to q'', tz)}\ \text{Reduce}(A \to \omega)$$

(xv) $q_f := \mathrm{GOTO}(q_0, S)$.
(xvi) $L(LRA(G)) := \{z \in T^* \mid (\varepsilon : q_0 \to q_0, z\$) \vdash^+ (S : q_0 \to q_f, \$)\}$. $L(G) = L(LRA(G))$.
(xvii) For $ts_1 \subseteq \{z \in (T')^* \mid |z| \leq 1\}$ and $ts_2 \subseteq \{z \in (T')^* \mid |z| \leq 1\}$,

$$ts_1 \odot ts_2 := \begin{cases} ts_1, & \text{if } \varepsilon \notin ts_1; \\ (ts_1 \setminus \{\varepsilon\}) \cup ts_2, & \text{otherwise.} \end{cases}$$

(xviii) $\mathrm{First}(A) := \{t \mid t \in T', \omega \in V^*, A \Rightarrow_{P'}^* t\omega\} \cup \{\varepsilon \mid A \Rightarrow_{P'}^* \varepsilon\}$ for $A \in N'$.
(xix) $\mathrm{First}(t) := \{t\}$ for $t \in T'$.
(xx) $\mathrm{First}(\varepsilon) := \{\varepsilon\}$. $\mathrm{First}(X\alpha) := \mathrm{First}(X) \odot \mathrm{First}(\alpha)$ for $X \in V'$ and $\alpha \in (V')^*$.

Theorem. (LA) $[A \to \omega\cdot] \in q$인 $q \in Q$, $A \in N'$와 $\omega \in V^*$에 대하여,

$$\mathbf{LA}(q, [A \to \omega]) \triangleq \left\{ t \in T' \,\middle|\, \begin{array}{c} (\alpha\omega : q_0 \to q, tz) \vdash_{\mathrm{Reduce}(A \to \omega)} (\alpha A : q_0 \to q', tz) \\ \&\& \\ (\alpha A : q_0 \to q', tz) \vdash^* (S : q_0 \to q_f, \$) \end{array} \right\}. \tag{1}$$

라 하자. 그러면

$$\mathbf{LA}(q, [A \to \omega]) = \left\{ t \in T' \,\middle|\, S' \underset{\mathbf{rm}\ P'}{\Longrightarrow}^* \alpha A z, \alpha\omega : q_0 \to q, \alpha \in V^*, z \in T^*, t \in \mathrm{First}(z\$) \right\} \tag{2}$$

이 성립한다.

이제 $\mathbf{LA}' : \{(q, [A \to \alpha \cdot \beta]) \mid q \in Q, [A \to \alpha \cdot \beta] \in q\} \to \mathcal{P}(\{z \in (T')^* \mid |z| \leq 1\})$를

$$\mathbf{LA}' =_\mu \mathbf{propagate}_{\mathbf{LA}'}, \tag{3}$$

로 정의하자. 여기서 **propagate**의 도입 규칙은 모두 3가지로

$$\frac{}{\$ \in \mathbf{propagate}_{\mathbf{LA}'_{\text{previous}}}(q_0, [S' \to \cdot S])}\ \textbf{Initial} \tag{4}$$

$$\frac{}{\$ \in \mathbf{propagate}_{\mathbf{LA}'_{\text{previous}}}(q_f, [S' \to S \cdot])}\ \textbf{Final} \tag{5}$$

$$\frac{q \in Q \quad [A \to \alpha \cdot \beta] \in q \quad [A \to \alpha\beta] \in P \quad \alpha : p \to q \quad [B \to \alpha_1 \cdot A\alpha_2] \in p}{\mathrm{First}(\alpha_2) \odot \mathbf{LA}'_{\text{previous}}(p, [B \to \alpha_1 \cdot A\alpha_2]) \subseteq \mathbf{propagate}_{\mathbf{LA}'_{\text{previous}}}(q, [A \to \alpha \cdot \beta])}\ \textbf{Step} \tag{6}$$

이다. 그러면, $q \in Q$이고 $[A \to \omega \cdot] \in q$일 때마다,

$$\mathbf{LA}(q, [A \to \omega]) = \{t \in T' \mid t \in \mathbf{LA}'(q, [A \to \omega \cdot])\} \tag{7}$$

이 성립한다. 이는

$$\mathbf{LA}'(q, [A \to \alpha \cdot \beta]) = \bigcup \left\{ \mathrm{First}(z\$) \;\middle|\; S' \underset{\mathbf{rm}\ P'}{\Longrightarrow}^* \omega A z \underset{\mathbf{rm}\ P'}{\Longrightarrow} \omega\alpha\beta z, \omega\alpha : q_0 \to q, z \in T^* \right\} \tag{8}$$

이기 때문이다.

Remark. (Fixed-point) 완비격자 $(D, \leq)$가 주어졌다고 하자. 함수

$$f : D \to D$$

가 단조증가할 때, $x_0 \in D$ 이상의 f의 고정점 중 가장 작은 것을 구하는 법을 생각해보자.

$$\begin{aligned} x : \mathsf{Ord} &\to D \\ S(\alpha) &\mapsto x_0 \vee f(x_\alpha) \\ \alpha &\mapsto \bigvee_{\zeta < \alpha} x_\zeta \quad \text{if } \alpha \in \mathsf{LimOrd}_{>0} \end{aligned}$$

로 잡고, $\kappa := \mathbf{h}(D)$로 놓으면, x_κ가 구하는 그것이 된다. 즉,

$$x_\kappa = f(x_\kappa) \tag{9}$$

이고

$$(\forall x \in D)(x = f(x) \to (x_0 \leq x \leftrightarrow x_\kappa \leq x)) \tag{10}$$

이다. 여기서, 임의의 집합 A에 대하여

$$\mathbf{h}(A) \triangleq \{\alpha \in \mathsf{Ord} \mid \exists\, \iota : \alpha \hookrightarrow A\}$$

이며, A가 유한하면 $\mathbf{h}(A) = |A| + 1$이다.